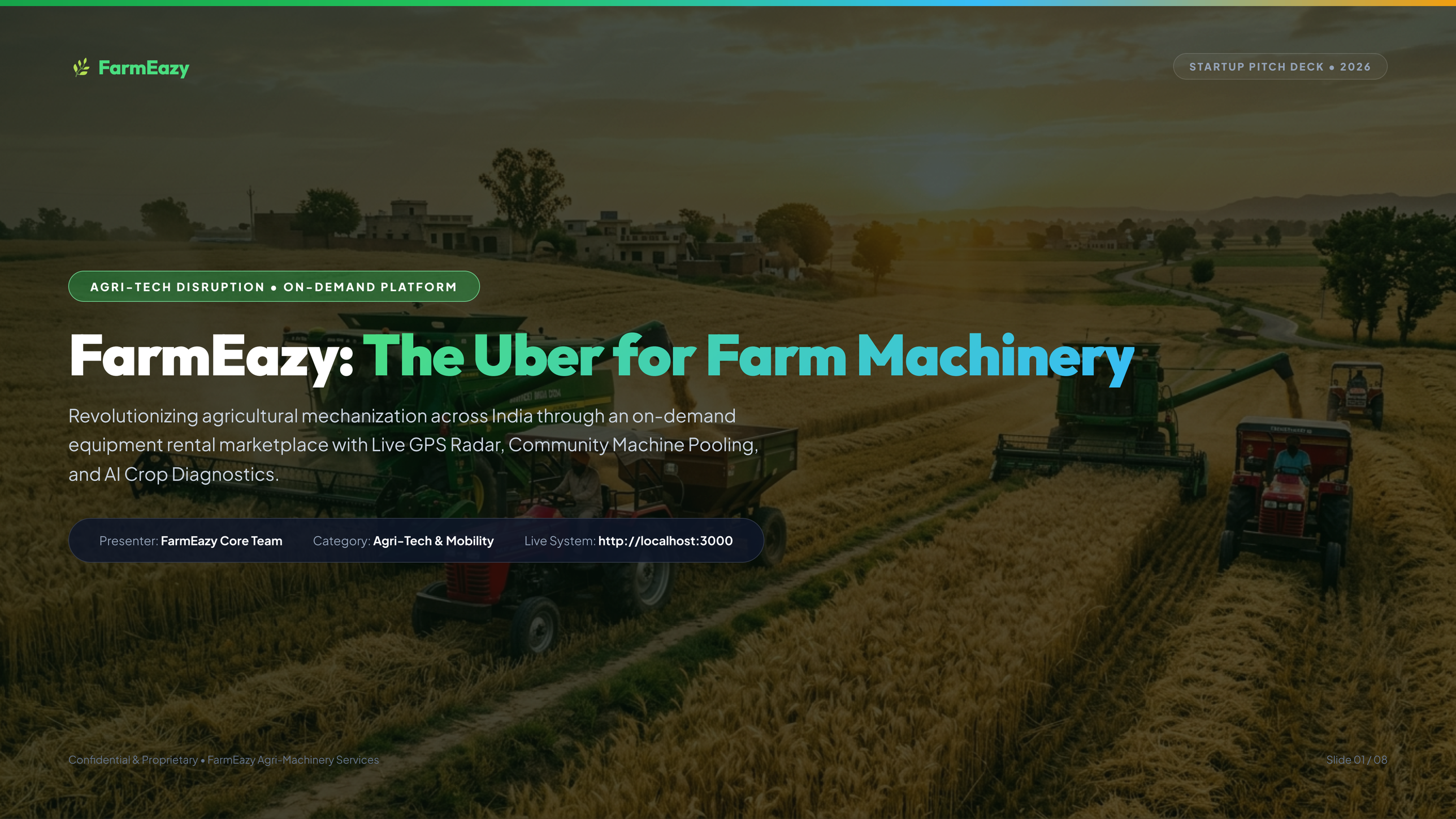


AGRI-TECH DISRUPTION • ON-DEMAND PLATFORM

FarmEazy: The Uber for Farm Machinery

Revolutionizing agricultural mechanization across India through an on-demand equipment rental marketplace with Live GPS Radar, Community Machine Pooling, and AI Crop Diagnostics.

Presenter: **FarmEazy Core Team** Category: **Agri-Tech & Mobility** Live System: **http://localhost:3000**

The Ground Reality: Critical Farming Bottlenecks

Over 86% of Indian farmers are smallholders trapped between high capital machinery costs and seasonal labor shortages.



Prohibitive Capital Cost

Modern tractors and harvesters cost ₹6.5 Lakhs to ₹15 Lakhs. Smallholder farmers cannot buy heavy equipment that remains unutilized for 8 to 9 months every year.



Peak Seasonal Shortages

During the critical 7–10 day sowing and harvesting windows, local village equipment gets booked out, causing crop spoilage, exorbitant rates, and huge yield losses.



Equipment Underutilization

Over 40% of India's 1.2M+ tractors sit idle outside peak cycles. Owners lack a digital platform to find nearby farmers who need tillage, spraying, or harvesting.

The FarmEazy Dual-Sided Marketplace

Connecting farmers needing equipment with machine owners seeking steady income — just like Uber.

For Farmers (Customer App)

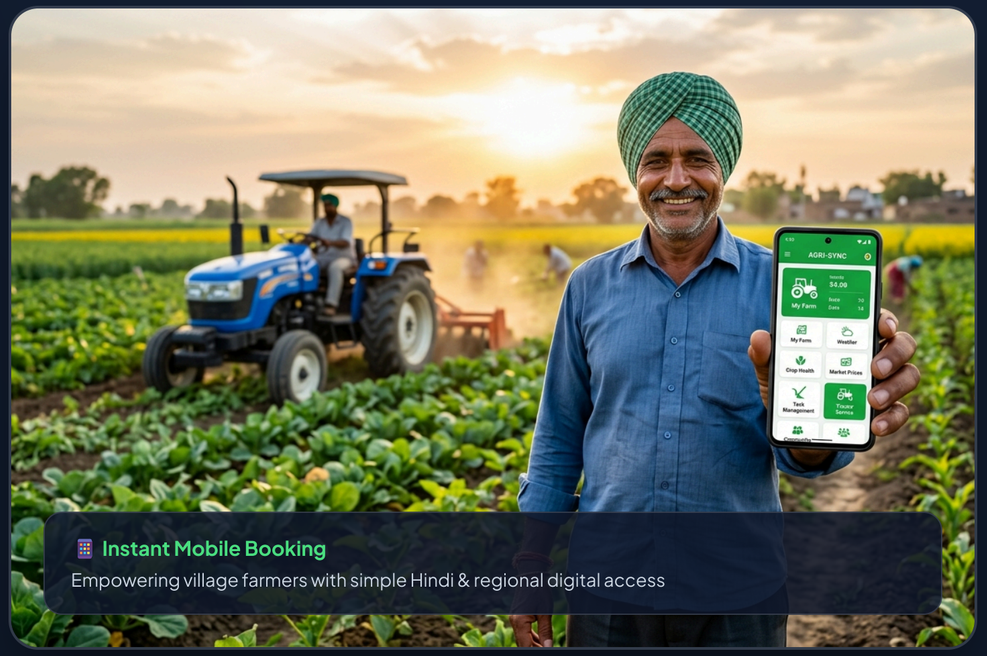
Zero Middlemen • Transparent Pricing

Search, compare, and book tractors by horsepower, attachments (rotavator, laser leveler, harvester), and hourly/daily rates with verified trained operators.

For Equipment Owners (Driver / Partner Hub)

Earnings Dashboard • Live Trip Management

Receive incoming field requests, 1-click order acceptance, route navigation to farms, and earn up to ₹38,000+ extra per season with guaranteed payment.



Instant Mobile Booking

Empowering village farmers with simple Hindi & regional digital access

4 Flagship High-Tech Innovations

Going far beyond simple listings with modern geospatial radar, agronomic calculations, and AI diagnostics.



Live GPS Radar

Uber-style animated satellite map showing moving tractor coordinates, remaining distance, speed, and real-time arrival countdown.



Smart Fuel Estimator

Enter crop, acreage, and soil texture (sandy, clay, black soil) to calculate exact machine hours, diesel litres, and net labor savings.



Farm Pooling (UberPool)

Neighboring farmers club adjacent fields to share mobilization overhead, saving up to 25% on combine harvesters and rotavators.



AI Crop Doctor

Select leaf symptoms (Yellow Rust, Stem Borers) to get instant chemical dosage guidance and recommended precision sprayers.

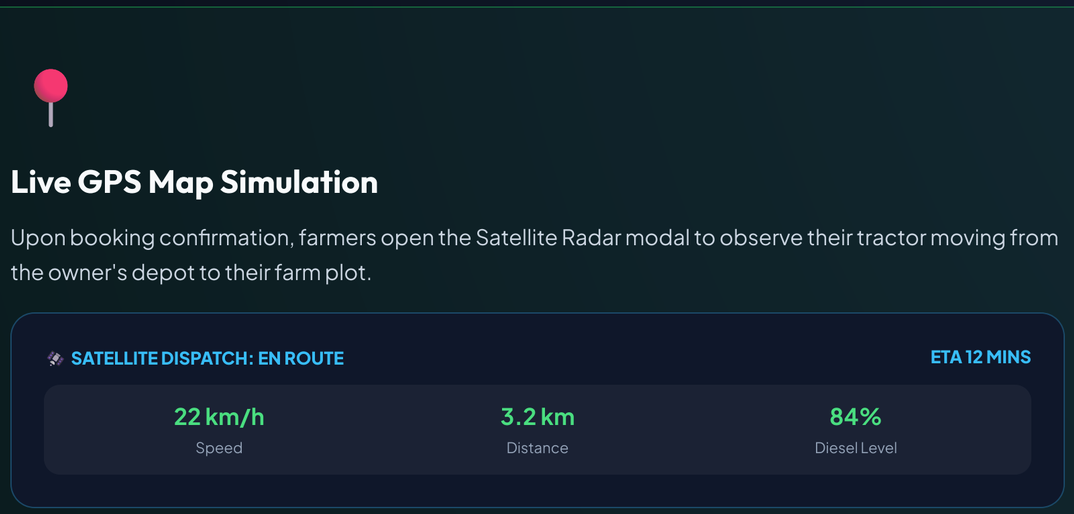
Deep Dive: Live GPS Radar & Driver Operations

Solving the rural trust barrier through real-time telemetry and structured order workflow states.

📍

Live GPS Map Simulation

Upon booking confirmation, farmers open the Satellite Radar modal to observe their tractor moving from the owner's depot to their farm plot.



👩‍🌾

Driver Order Lifecycle Pipeline

Equipment operators oversee all village bookings with status actions directly synced to the SQLite backend:

1. **Confirmed →** Automatic driver dispatch alert
2. **Accepted →** Operator confirms availability
3. **On The Way →** Real-time GPS radar active
4. **Work In Progress →** Field tillage / harvesting underway
5. **Completed →** Field verified & payment settled

Full-Stack System Architecture

A resilient, responsive, and secure tech stack designed for fast loading even on rural 4G connections.

FRONTEND	Vanilla HTML5, Glassmorphism CSS design system, Responsive layout (desktop to smartphone), Vanilla JavaScript single-page controllers.
BACKEND SERVER	Node.js with Express.js REST API, Helmet security hardening, express-session cookie authentication, and rate limiting.
PERSISTENT DATABASE	Relational SQLite (SQL.js) with isolated storage (`database/farmeazy.sqlite`), bcrypt credential hashing, and foreign key role separation.
AGRI-INTELLIGENCE	APMC Mandi rate engine, real-time agro-weather alerts, algorithmic diesel estimators, and rule-based diagnostic crop inference.

Measurable Socio-Economic Impact

Empowering smallholder farmers with higher productivity while increasing machinery asset yields.

30%
Tillage & Sowing Cost Reduction

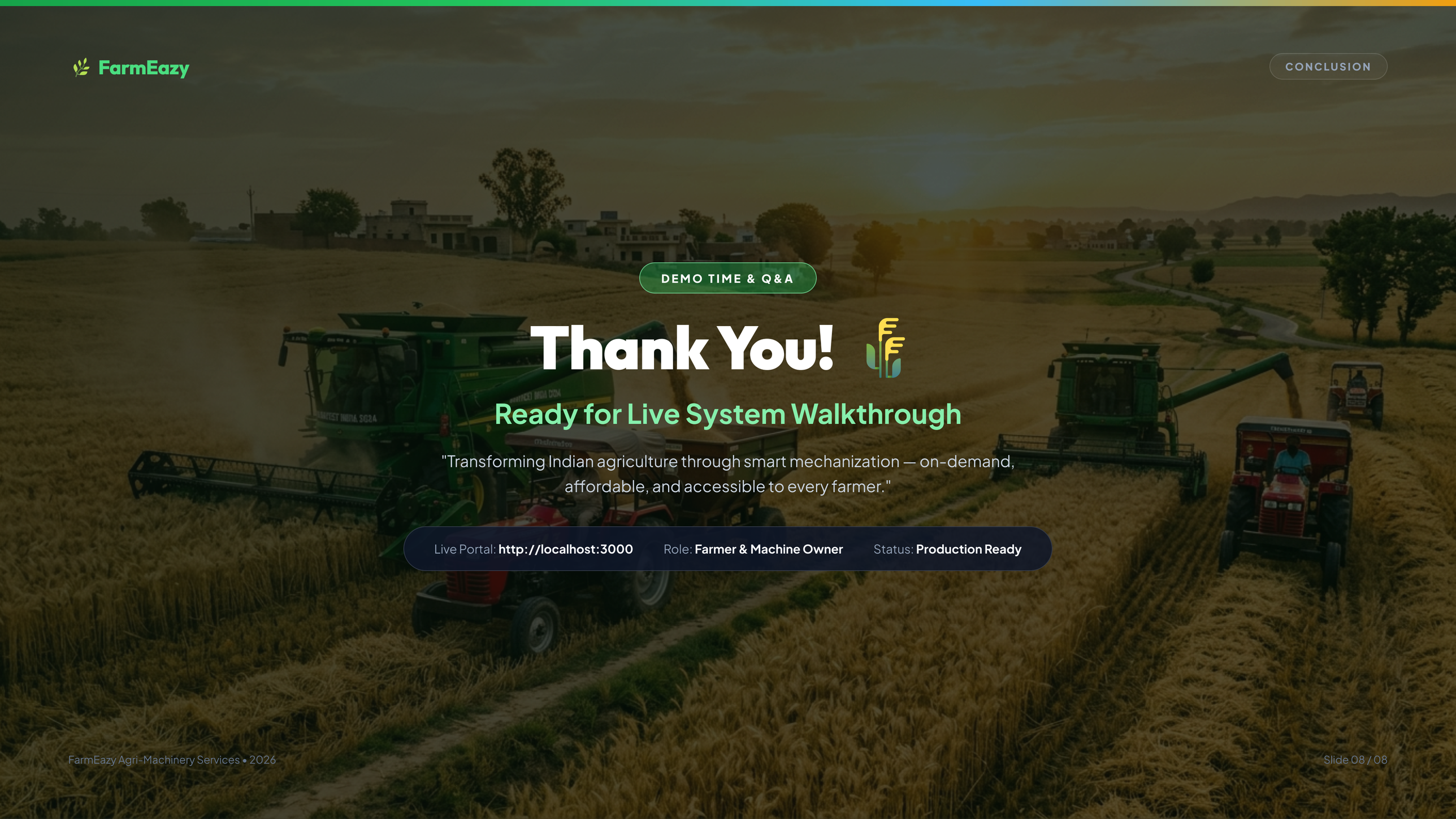
15 Min
Average Booking Response Time

₹38,400
Average Monthly Machine Owner Revenue

100%
Verified Drivers & Field Quality Check

🚀 Scalability Roadmap

Upcoming milestones include automated UPI escrow payments, IoT fuel tank sensors for anti-theft monitoring, vernacular voice-based bookings (Hindi, Punjabi, Haryanvi), and satellite NDVI crop yield mapping.



DEMO TIME & Q&A

Thank You!

Ready for Live System Walkthrough

"Transforming Indian agriculture through smart mechanization — on-demand, affordable, and accessible to every farmer."

Live Portal: **http://localhost:3000** Role: **Farmer & Machine Owner** Status: **Production Ready**